

Credit-Linked Note (CLN) Pricer

model: "cln"

Reduced-form hazard-rate model on CDS curve

Description

Prices a credit-linked note as a risky bond whose coupons and redemption are weighted by the survival probability of the reference entity. The hazard rate term structure is calibrated to CDS spreads from the curve file. A recovery-rate-weighted default leg adds the expected present value of recoveries. Yields are computed both on a promised basis (ignoring default) and on an expected basis (weighted by survival).

Pricing Methodology

CDS spreads are converted to piecewise-constant hazard rates:

$$\lambda(t) = s(t) / (1 - R)$$

Survival probability:

$$S(t) = \exp(-\int_0^t \lambda(u) du)$$

Default probability in interval $[t_{i-1}, t_i]$: $\Delta P = S(t_{i-1}) - S(t_i)$

Key Formulas

$$\begin{aligned} NPV &= \sum \text{coupon}(t) \cdot S(t) \cdot DF(t) \\ &+ \text{par} \cdot S(T) \cdot DF(T) \\ &+ R \cdot \text{par} \cdot \sum \Delta P(t) \cdot DF_{\text{avg}}(t) \\ \lambda(t) &= s(t) / (1 - R) && \text{[hazard from CDS spread]} \\ S(t) &= \exp(-\sum \lambda_i \cdot \Delta t_i) && \text{[piecewise-constant } \lambda \text{]} \end{aligned}$$

Required Fields

Field	Type	Description
instrument_id	string	ISIN or internal identifier
evaluation_date	string	Pricing date
issue_date	string	Issue date
maturity_date	string	Maturity date
fixed_coupon_rate	float	Annual coupon rate on the CLN
coupon_frequency	string	Annual Semiannual Quarterly Monthly
accrual_day_count	string	Day count convention
par	float	Face value (e.g. 100)
reference_entity	string	Name of reference entity (for CDS curve lookup)

Optional Fields

Field	Type	Description
recovery_rate	float	Default recovery rate (default 0.40)
credit_spread_bp	float	Issuer z-spread over risk-free rate (default 0)
reference_cds_curve_name	string	Explicit CDS curve override

Key Outputs

- selected_npv — total CLN NPV
- pv_coupons, pv_redemption, pv_recovery — component breakdown
- ytm_promised — yield ignoring default risk
- ytm_expected (= ytm) — yield weighted by survival / recovery
- hazard_segments — calibrated λ per tenor

- `cds_curve_used` — which CDS curve was selected